

BLACKJACK OUTCOME ANALYSIS

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MOTIVATION & GOALS

Our motivation is to explore how data-driven methods can reveal meaningful patterns in blackjack outcomes. By analyzing a large real-world dataset, we aim to apply and strengthen the data-mining skills learned in the IDS2025 course.

- Goal 1: Discover frequent patterns in winning and losing hands.
- Goal 2: Predict the outcome of each hand using machine learning models (decision tree, k-NN, random forest) by analyzing pre-decision features
- Goal 3: Evaluate model performance

DATA ANALYSIS

We used the Kaggle - 900,000 Hands of BlackJack Results dataset. Firstly we analyzed only the pre-decision features (player's initial cards and dealers upcard) to understand early-game blackjack patterns. Outcome frequencies reveal that losing is more common than winning (matches with blackjack expectations). Dealer upcard analysis confirms that players have their best winning chances when the dealer shows a low card 2-6, and low win chances when the dealer shows 7-11. Correlation analysis shows that the player's starting total is the strongest predictor of winning. The dealer upcard has a moderate negative correlation. Soft hands occur in a small portion of hands and have lower early win rates, as they often represent flexible but weaker starting positions. Overall, the exploratory analysis highlights stable and logical patterns that align with known blackjack strategy principles, validating the dataset and supporting the next steps in feature engineering and model development.

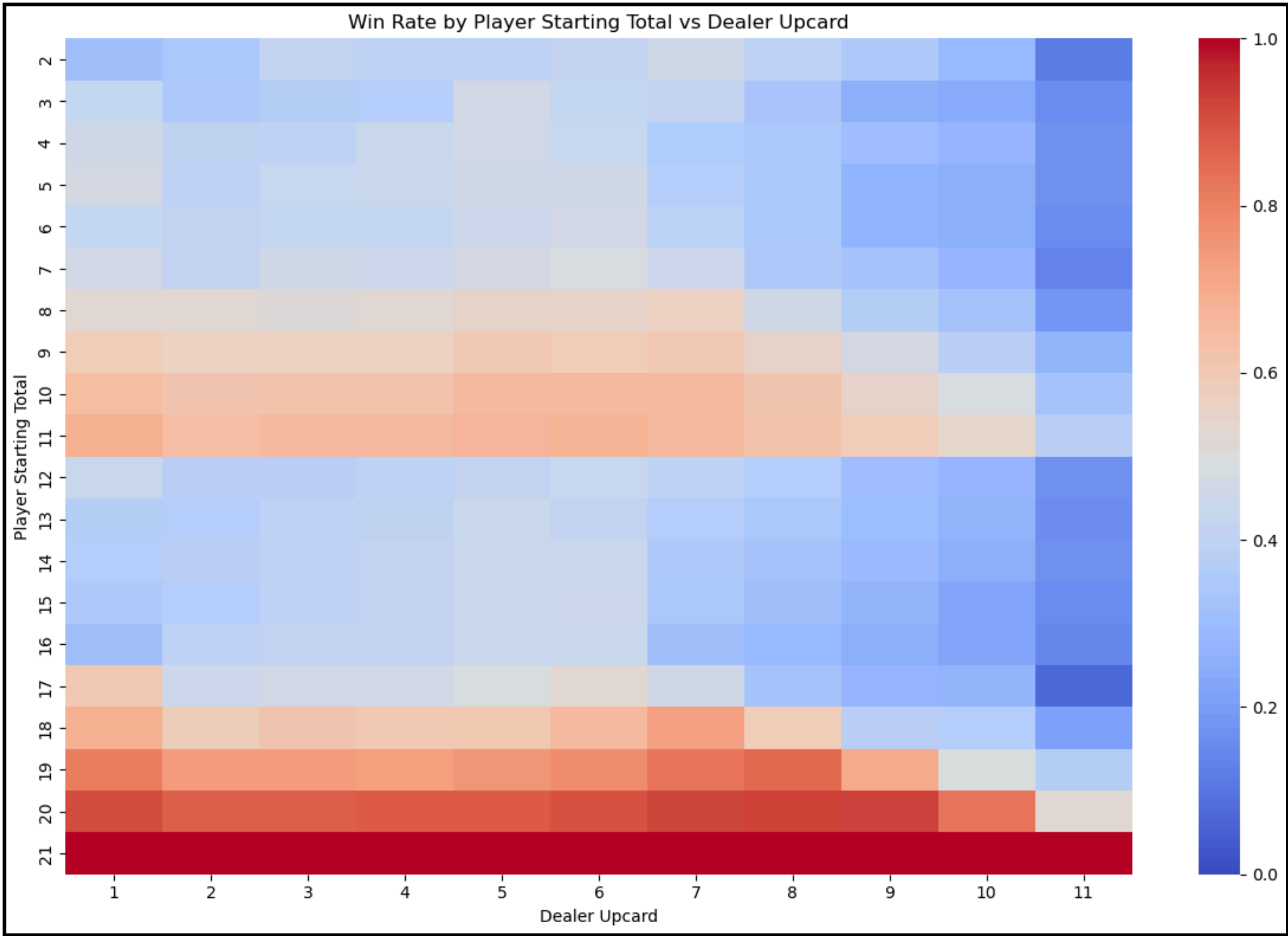
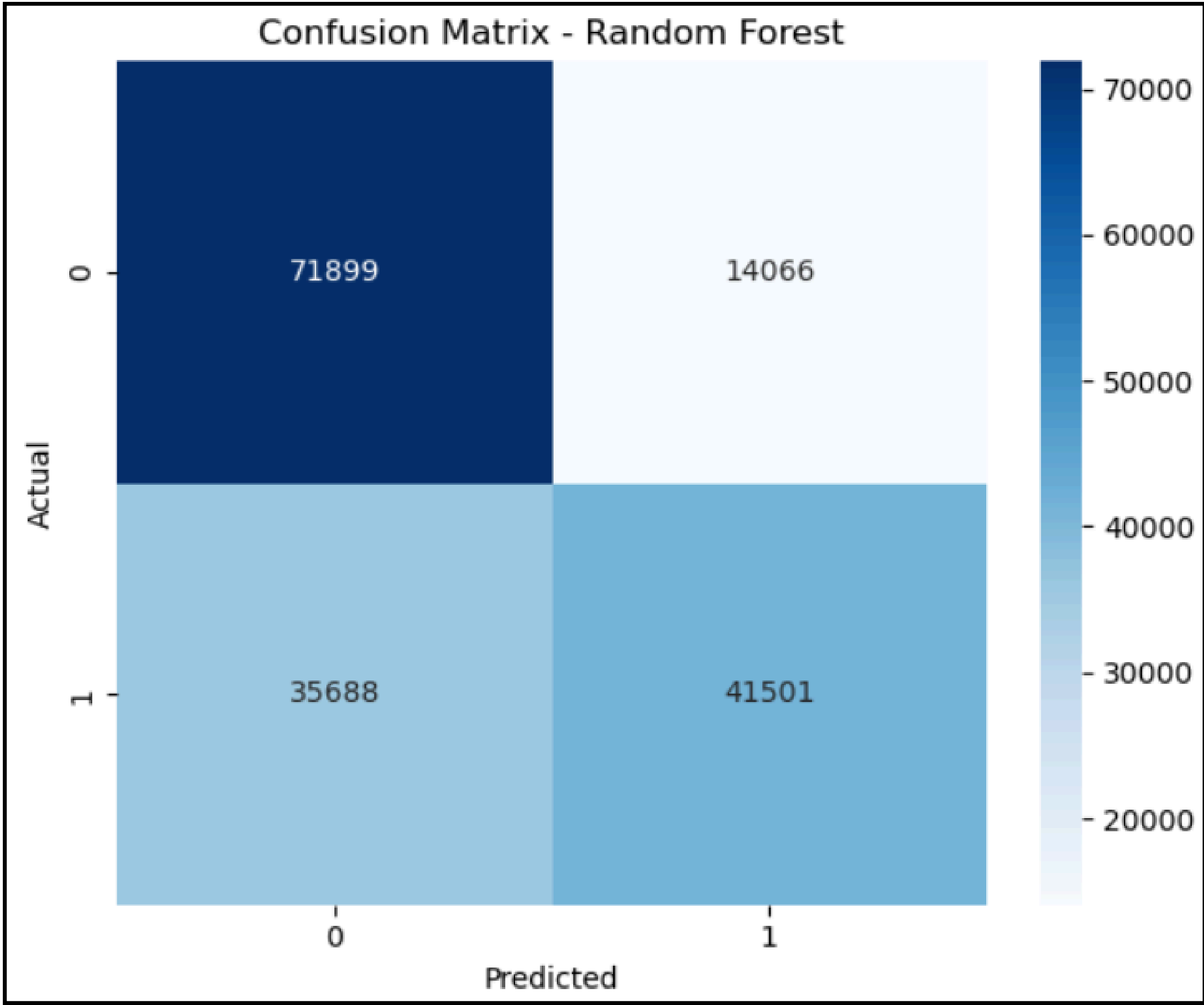


Table 1: Feature Engineering

card1	card2	dealcard1	winloss	player_total_pre	is_soft_hand	has_ace	is_pair	dealer_strength	player_risk_zone	advantage_metric
11	8	8	1	19	0	0	0	neutral	high	11
1	1	8	0	2	1	1	1	neutral	low	-6
8	10	11	0	18	0	0	0	strong	high	7
2	9	10	1	11	0	0	0	strong	low	1
10	10	10	1	20	0	0	1	strong	high	10

MODEL DEVELOPMENT

Firstly we did feature engineering to provide the models with meaningful structure beyond the basic card values, such as player risk zones and an advantage metric (Table 1). Using this enhanced dataset, we trained several machine learning models to predict hand outcomes before any player actions occur. Models included Logistic Regression, Decision Tree, Random Forest, k-Nearest Neighbors, Support Vector Machine, Gradient Boosting, and LightGBM. Random Forest and LightGBM achieved the strongest performance at approximately 0.69 accuracy, which is strong given the inherent randomness of blackjack. We also performed hyperparameter tuning with RandomizedSearchCV on a 50,000-hand subsample to efficiently optimize model settings.



RESULTS & EVALUATION

Feature importance analysis showed that player starting total is the strongest predictor of winning, with the dealer upcard and the advantage metric also ranking highly. These insights satisfy our first goal by revealing the most frequent and meaningful patterns behind win and loss outcomes. Our best predictive model was the Random Forest, achieving approximately **0.69** accuracy, demonstrating that machine learning can successfully estimate hand outcomes using only pre-decision information. This meets our second goal of building outcome-prediction models. Finally, the evaluation confirms that the combination of solid model accuracy and interpretable feature rankings provides a reliable foundation for data-driven strategy insights, addressing our third goal of assessing model performance.

REFERENCES

- 1.Kaggle dataset: <https://www.kaggle.com/datasets/mojocolors/900000-hands-of-blackjack-results>
- 2.Blackjack rules: <https://bicyclecards.com/how-to-play/blackjack>